THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The number of factors of 72 is: 1

(a) 13 **(b)** 14 **(c)** 12 **(d)** 11

2. A group of 12 people has average weight 71 kg. Another group has average weight 74 kg. Together the average weight is 73 kg. The number of people in the second group is: 1

(a) 24 **(b)** 25 **(c)** 26 **(d)** 23

3. Set s = 0. While n > 0: add (n mod 10) to s, then replace n by the quotient of n divided by 10. Starting with n = 32481, the final s is: 1

(a) 18 **(b)** 19 **(c)** 17 **(d)** 20

4. The diagonals of a rhombus are 10 cm and 24 cm. Its perimeter is: 1

(a) 56 cm **(b)** 65 cm **(c)** 78 cm **(d)** 52 cm

5. The line $3x + 2y = 6$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 5 **(b)** 2 **(c)** 3 **(d)** 4

6. Take $\pi = \frac{22}{7}$. The total surface area of a cone of radius 10.5 cm and slant height 17.5 cm is: 1

(a) 1110 cm^2 **(b)** 1017 cm^2 **(c)** 831 cm^2 **(d)** 924 cm^2

7. If $ab > 0$ and $a + b < 0$, the point (a, b) lies in: 1

(a) Quadrant IV **(b)** Quadrant II **(c)** Quadrant III **(d)** Quadrant I

8. If $p(x) = 5x + 2$ and $q(x) = 3x - 7$, then $p(x) + q(x) =$ 1

(a) $9x - 5$ **(b)** $8x - 5$ **(c)** $8x - 6$ **(d)** $8x - 4$

9. $\sqrt{3} \times \sqrt{12} =$ 1

(a) 7 **(b)** 5 **(c)** 8 **(d)** 6

10. The value of 1003^2 is: 1

(a) 1006010 **(b)** 1006011 **(c)** 1006009 **(d)** 1006008

11. $\angle AOB = 96^\circ$ at the centre and C lies on the major arc AB. Then $\angle ACB =$ 1

(a) 38° **(b)** 68° **(c)** 58° **(d)** 48°

12. The area of a quadrilateral whose diagonal is 20 cm and whose offsets from the diagonal are 6 cm and 8 cm is (cm^2): 1

(a) 139 **(b)** 141 **(c)** 140 **(d)** 142

13. A letter is chosen at random from the word MISSISSIPPI. The probability that it is the letter S is: 1

(a) $\frac{4}{11}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{5}{11}$ **(d)** $\frac{7}{11}$

14. The sum of the first 14 odd natural numbers is: 1

(a) 224 **(b)** 182 **(c)** 210 **(d)** 196

15. A triangle has sides 33 cm, 56 cm and 65 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) not right-angled **(b)** equilateral **(c)** isosceles **(d)** right-angled

16. 360 mL of a 35% sugar solution is mixed with 680 mL of a 9% sugar solution. The concentration of the mixture is: 1

(a) 17% **(b)** 20% **(c)** 18% **(d)** 19%

17. Using $\gcd(20,\,25) = 5$, the LCM of 20 and 25 is: 1

(a) 105 **(b)** 95 **(c)** 120 **(d)** 100

18. The diagonals of a rhombus are 10 cm and 24 cm. Its area is: 1

(a) 120 cm^2 **(b)** 144 cm^2 **(c)** 130 cm^2 **(d)** 110 cm^2

19. A taxi fare is ₹83 for 5 km and ₹193 for 15 km (fare is a linear function of distance). The fare for 21 km (in ₹) is: 1

(a) 270 **(b)** 248 **(c)** 287 **(d)** 259

20. A cone has radius 8 cm and height 15 cm. Its slant height is: 1

(a) 16 cm **(b)** 19 cm **(c)** 17 cm **(d)** 18 cm

21. The perimeter of the triangle with vertices $(0,0)$, $(9,0)$ and $(0,12)$ is: 1

(a) 38 **(b)** 37 **(c)** 36 **(d)** 35

22. A linear polynomial $p(x) = ax + b$ has $p(1) = 11$ and $p(2) = 18$. Then $b =$ 1

(a) 4 **(b)** 3 **(c)** 6 **(d)** 5

23. $\sqrt{3} + (5 - \sqrt{3})$ is: 1

(a) zero **(b)** negative **(c)** rational **(d)** irrational

24. The value of $78^2 - 22^2$ is: 1

(a) 5599 **(b)** 5601 **(c)** 5602 **(d)** 5600

25. $\angle AOB = 90^\circ$ at the centre and D lies on the minor arc AB. Then $\angle ADB =$ 1

(a) 135° **(b)** 145° **(c)** 125° **(d)** 155°

26. A rectangle of perimeter 50 cm has length 18 cm. Its area (cm^2) is: 1

(a) 125 **(b)** 127 **(c)** 126 **(d)** 128

27. In a town of 12000 people the probability that a randomly chosen person owns a smartphone is $\frac{3}{5}$. The expected number of owners is: 1

(a) 7201 **(b)** 7199 **(c)** 7202 **(d)** 7200

28. The ratio of the 5th term to the 3rd term of a GP is 9. The common ratio (positive) is: 1

(a) 4 **(b)** 5 **(c)** 2 **(d)** 3

29. Which statement is true for every integer n? 1

(a) n^2 is even **(b)** $n(n+1)$ is even **(c)** $n+1$ is even **(d)** $n(n+1)$ is odd

30. Marks of 62, 78 and 79 are given weights 3, 4 and 4 respectively. The weighted average is: 1

(a) 75 **(b)** 74 **(c)** 76 **(d)** 73

31. To test whether 373 is prime, we check divisibility by 2, 3, 4, … up to $\lfloor\sqrt{373}\rfloor$. The last number checked is: 1

(a) 186 **(b)** 20 **(c)** 18 **(d)** 19

32. The diagonals of a quadrilateral are 15 cm and 28 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 129 cm **(b)** 64 cm **(c)** 88 cm **(d)** 43 cm

33. The point $(-4, k)$ lies on the line $2x + 6y = 28$. The value of k is: 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

34. A tank measures 9 m $\times$ 7 m $\times$ 11 m. Its capacity in litres ($1\ \text{m}^3 = 1000$ L) is: 1

(a) 693 **(b)** 693000 **(c)** 6930000 **(d)** 69300

35. The point $(4, 9)$ is reflected first in the x-axis and then in the y-axis. The final point is: 1

(a) $(-4, -9)$ **(b)** $(-4, 9)$ **(c)** $(-9, -4)$ **(d)** $(4, -9)$

36. The nth term of a pattern is $7n - 2$. Its 12th term is: 1

(a) 82 **(b)** 81 **(c)** 84 **(d)** 83

37. A rational number between $\frac{2}{7}$ and $\frac{4}{7}$ is: 1

(a) $\frac{1}{7}$ **(b)** $\frac{5}{7}$ **(c)** $\frac{3}{7}$ **(d)** $\frac{6}{7}$

38. Factorise $x^2 - 3x - 28$: 1

(a) $(x - 7)(x - 4)$ **(b)** $(x - 7)(x + 4)$ **(c)** $(x - 6)(x + 5)$ **(d)** $(x + 7)(x - 4)$

39. In a circle, a chord of length 18 cm is at distance 12 cm from the centre. Another chord of the same length is at a distance (in cm): 1

(a) 11 **(b)** 14 **(c)** 13 **(d)** 12

40. A circular wire of radius 7 cm (take $\pi = \frac{22}{7}$) is bent into a square. The side of the square (cm) is: 1

(a) 12 **(b)** 10 **(c)** 11 **(d)** 13

41. If $P(E) = 5/12$, then the probability that E does not happen is: 1

(a) $\frac{7}{13}$ **(b)** $\frac{7}{12}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{5}{12}$

42. The number of multiples of 7 between 20 and 150 (both excluded) is: 1

(a) 20 **(b)** 18 **(c)** 21 **(d)** 19

43. The sum of the 5 consecutive integers starting from 17 is: 1

(a) 90 **(b)** 95 **(c)** 100 **(d)** 96

44. The average monthly salary of 14 employees is ₹56 thousand. An employee earning ₹28 thousand leaves and a new employee joins, and the average rises by ₹3 thousand. The new employee's salary (in ₹ thousand) is: 1

(a) 75 **(b)** 70 **(c)** 67 **(d)** 73

45. Euclid's subtraction algorithm is applied to find gcd$(312, 200)$. The number of subtractions performed (until a pair with 0 appears) is: 1

(a) 9 **(b)** 10 **(c)** 8 **(d)** 11

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46. In a parallelogram, two opposite angles are $(3x + 17)^\circ$ and $(x + 105)^\circ$. The value of x is: 1

(a) 44 **(b)** 46 **(c)** 45 **(d)** 43

47. Which of the following points lies on the line $8x + 5y = 27$? 1

(a) $(7, -1)$ **(b)** $(0, 7)$ **(c)** $(-1, 7)$ **(d)** $(-1, 8)$

48. The number of cubes of edge 5 cm that can be cut from a cuboid 30 cm $\times$ 15 cm $\times$ 20 cm is: 1

(a) 71 **(b)** 72 **(c)** 78 **(d)** 73

49. The area of the rectangle with opposite vertices $(1, 2)$ and $(6, 5)$ and sides parallel to the axes is: 1

(a) 16 **(b)** 15 **(c)** 17 **(d)** 14

50. Lines $y = 3x + 2$ and $y = 3x - 7$ are: 1

(a) perpendicular **(b)** intersecting at one point
(c) parallel **(d)** identical

END OF QUESTION PAPER **Total: 50 Marks**

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